Syed Muhammad Mehdi Abidi

+92-332-3488793 mehdiali78666@gmail.com linkedin.com/in/mehdiabidii
github.com/Mehdi-Abidi

Education

- **FAST NUCES, Karachi** – B.S. Computer Science, **CGPA: 3.85/4.0** (Exp. Jun 2026).
  - **Achievements**: **4x Gold Medalist**, **Rector's Honor List (4x)**, **Dean's Honor List (3x)**.
  - **Coursework**: Deep Learning, AI, Recommender Systems, Generative AI, Probability & Statistics.

Research & Final Year Project

SecureSphere-LLM: AI Middleware for Secure LLM Usage

- Designed a security-focused middleware to intercept and inspect prompts sent from developer tools (VS Code) to LLMs (capable of parsing requests from any LLM behind GitHub Copilot) using a Node.js MITM proxy running as a background system service.
- Implemented layered guardrails including regex- and heuristic-based detection for prompt injection and jailbreaks, Microsoft Presidio for PII leakage detection, and NVIDIA NeMo Guardrails for policy-based control.
- Added audit logging and rule-based enforcement to ensure compliance, traceability, and safe LLM usage, demonstrating a practical **LLMSecOps** approach for enterprise environments.

Experience

- **AI AI Engineer Lead – Agento (Jun 2026–Present)**
  - Lead trustworthy-AI engineering for an enterprise agentic workflow-automation platform built on **FastAPI, Temporal durable workflows, React, PostgreSQL, and Azure**, where AI agents compose and execute multi-step business workflows end to end.
  - Operate as the **L2 reviewer** in the engineering review hierarchy; security-reviewed **110+ pull requests** covering **SSRF, IDOR, XSS, CSRF, ReDoS, header injection, and prompt-injection** vectors, gating production merges across backend and frontend.
  - Architected a multi-tenant **Knowledge Source (RAG) module**: PDF/DOCX/URL ingestion, **OpenAI embeddings**, **pgvector HNSW** semantic retrieval, and **PostgreSQL row-level security** for tenant isolation; delivered across **7 incremental releases with 29 automated tests**, including **SSRF-hardened** web ingestion via DNS-pinned fetching with private-address blocking and per-redirect revalidation.
  - Engineered **long-term memory** for AI agents with **Azure Blob**-backed snapshots and debounced refresh, enabling personalized cross-session context in agent responses.
  - Implemented **complexity-based LLM model routing** across **OpenAI, Anthropic, Moonshot, and Gemini**, directing low-complexity tasks to models up to **30x cheaper** per token while preserving output quality.
  - Hardened **6 public no-auth API endpoints** with spoof-resistant per-IP **rate limiting** and **Cloudflare Turnstile CAPTCHA**, and secured public webhook ingress with constant-time **HMAC** verification.
  - Optimized the AI **workflow builder** and production observability: **Azure Application Insights** telemetry with failure-only exception capture, breadcrumb tracing, and privately uploaded hidden sourcemaps, plus live-run reconnection and run-result UX improvements that reduced telemetry noise and debugging time.
- **AI Software Engineer 1 – Imperium Dynamics (Jan 2026–Jun 2026)**
  Built scalable **LLM applications** and **LangChain agents** using modular Python architectures. Developed AI-driven workflows and cloud automation with **Microsoft Fabric**, **Azure AI Foundry**, and **Azure Pipelines**. Utilized **Git**, **PDM**, and secure backend authentication for efficient development and deployment.
- **Associate Developer – Analytic Insider (Jan 2025–Jan 2026)**
  Built Generative AI applications including a **Tax Consultant Bot** using **RAG, LangChain, and Pinecone**, improving query-retrieval efficiency by **12%**. Automated structured data pipelines covering **3,000+ business directories** to power scalable LLM applications.
- **Data Science Intern – 10Pearls (Sep 2024–Dec 2024)**
  Developed an **AQI prediction system** using Random Forest, Ridge Classifier, and Logistic Regression with **87%** accuracy. Built real-time pipelines with the OpenWeather API & Hopsworks; automated CI/CD with GitHub Actions, cutting deployment time by **30%**.

Projects

- **PathoChat / PathoChat 2.0**: AI-powered medical chatbot using **LangChain, FAISS, Hugging Face, and Mistral-7B**, achieving **92% accuracy**. Extended into an agentic AI system with **n8n workflows** and **Pinecone** for scalable vector search.
- **Fingerprint Matcher**: Full-stack biometric recognition using **SIFT, PostgreSQL**, and a **Kotlin** mobile app; achieved **84%** real-time matching accuracy.

Skills

- **Programming & Backend**: Python, SQL, FastAPI, Django, Flask, MongoDB, PostgreSQL (pgvector, Row-Level Security).
- **ML/AI**: Scikit-learn, PyTorch, TensorFlow, Keras.
- **GenAI & Agents**: Hugging Face, LangChain, RAG, OpenAI API, Anthropic Claude, Pinecone, FAISS, n8n, Temporal Workflows, LLM Model Routing, LLM Guardrails (Input/Output Filtering, PII Redaction, Topical Safety, Prompt-Injection Defense).
- **Cloud & DevOps**: Azure (Blob Storage, Application Insights, AI Foundry, Fabric, Pipelines), Docker, GitHub Actions, CI/CD, Git.
- **Visualization**: Matplotlib, Plotly, Seaborn, SHAP, LIME.

Leadership

- **President – IEEE FAST-NUCES Student Branch** (2025–Present).
- **Co-Head DS/ML – GDSC and FDSS** (2024–2025)
  Led 10+ workshops, hackathons, and seminars engaging 500+ students.

Certifications

- **Convolutional Neural Networks**: DeepLearning.AI (Coursera, 2025)
  Credential ID: D055BIR477DK
- **Copilot Studio Agent Academy – Recruit**: Global AI Community (2026)
- **Microsoft Applied Skills Credential**: Microsoft Learn (2026)
- **Microsoft Applied Skills Credential**: Microsoft Learn (2026)
- **Developing LLM Applications with LangChain**: DataCamp
- **Large Language Models (LLMs) Concepts**: DataCamp
- **Generative AI Concepts**: DataCamp